Programming Assignment – Toward DES

Background

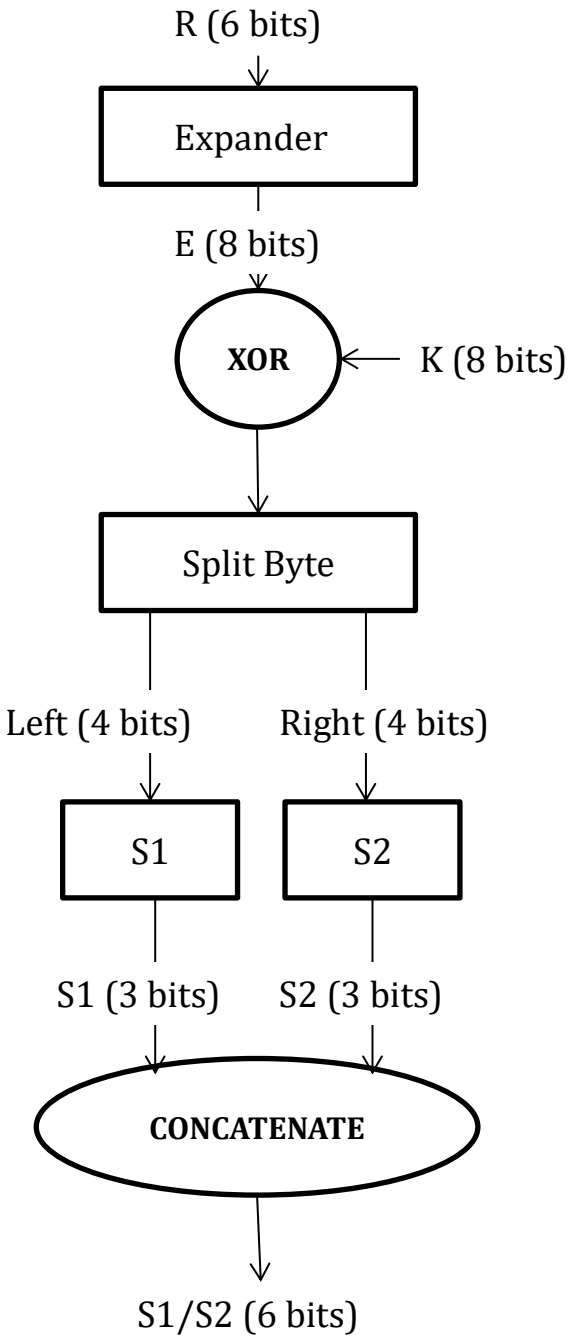
Advanced cryptography systems use various forms of substitution, transposition, and key-based arithmetic/logical operations to perform encryption and decryption. Many of those operations are rooted in the *Feistel* system, named after Horst Feistel, who was part of the IBM team that developed LUCIFER, the basis for the Data Encryption Standard (DES) cryptography system, which will be presented in an upcoming lecture. In this assignment you will implement a Feistel System function.

Specification

The Feistel System function you will implement receives two arguments, a 6-bit value and an 8-bit key. It returns a 6-bit value stored in the right-most 6 bits of an 8-bit **`byte`**. Your function should implement these as bytes using the following method definition:

```
public static byte feistel(byte R, byte K)
```

The operations are shown in the following flow chart:



Use the class **`BitOperationsPartIVTest.java`** (provided) to demonstrate your code. You may make naming changes to the entry point function (**`main`**) but you may not change the functionality.

Deliverables

- Source code (.java) files
- Reflective essay describing
    - Successes
    - Difficulties (if any) and how you addressed them
    - Lessons learned
    - Screen shot of your running program using the provided main